

The Hamiltonian is

$$\mathcal{H} = \sum_{\langle ij \rangle} J_{ij} \left(\vec{S}_i \cdot \vec{S}_j + \kappa \left(\vec{S}_i \cdot \vec{S}_j \right)^2 \right) \quad (1)$$

Using the Abrikosov representation of the spin in terms of fermions:

$$\vec{S}_i = \sum_{\alpha\beta} f_{i\alpha}^\dagger \frac{\vec{\sigma}}{2} f_{i\beta} \quad (2)$$

$$\begin{aligned} \Rightarrow \vec{S}_i \cdot \vec{S}_j &= \frac{1}{4} \sum_{\alpha\beta\gamma\delta} f_{i\alpha}^\dagger f_{i\beta} f_{j\gamma}^\dagger f_{j\delta} \underbrace{\vec{\sigma}_{\alpha\beta} \vec{\sigma}_{\gamma\delta}}_{=\delta_{\alpha\delta}\delta_{\beta\gamma} - \delta_{\alpha\gamma}\delta_{\beta\delta}} \\ &= \frac{1}{4} \sum_{\alpha\beta} 2f_{i\alpha}^\dagger f_{i\beta} f_{j\beta}^\dagger f_{j\alpha} - f_{i\alpha}^\dagger f_{i\alpha} f_{j\beta}^\dagger f_{j\beta} \end{aligned}$$

Move the last annihilation operator in the first term to the second position: use $f_{j\beta}^\dagger f_{j\alpha} = \delta_{\alpha\beta} - f_{j\alpha} f_{j\beta}^\dagger$ and acquire 1 minus signs. After that bring the other annihilation operator to the last position which gives another minus.

$$\vec{S}_i \cdot \vec{S}_j = \frac{1}{4} \sum_{\alpha\beta} -2f_{i\alpha}^\dagger f_{j\alpha} f_{j\beta}^\dagger f_{j\beta} + 2f_{i\alpha} f_{i\beta} \delta_{\alpha\beta} - f_{i\alpha}^\dagger f_{i\alpha} f_{j\beta}^\dagger f_{j\beta} \quad (3)$$

The second term is n_i and the third is $n_i n_j$. For the first term I introduce the shorthand

$$\chi_{ij} = \frac{1}{2} \sum_{\alpha} f_{i\alpha}^\dagger f_{j\alpha} \quad (4)$$

$$\vec{S}_i \cdot \vec{S}_j = -2\chi_{ij}\chi_{ji} + \frac{1}{2}n_i - \frac{1}{4}n_i n_j \quad (5)$$

Using the constraint $n_i = n_j = 1$, the final expression is

$$\vec{S}_i \cdot \vec{S}_j = -2\chi_{ij}\chi_{ji} + \frac{1}{4} \quad (6)$$

Relevant for the Hamiltonian is also the term:

$$\Rightarrow \left(\vec{S}_i \cdot \vec{S}_j \right)^2 = \frac{1}{16} - \chi_{ij}\chi_{ji} + 4(\chi_{ij}\chi_{ji})^2 \quad (7)$$

The mean field decoupling works like

$$\chi_{ij}\chi_{ji} \approx \langle \chi_{ij} \rangle \chi_{ji} + \langle \chi_{ji} \rangle \chi_{ij} - |\langle \chi_{ij} \rangle|^2 \quad (8)$$

$$(\chi_{ij}\chi_{ji})^2 \approx 2|\langle \chi_{ij} \rangle|^2 \langle \chi_{ij} \rangle \chi_{ji} + 2|\langle \chi_{ij} \rangle|^2 \langle \chi_{ji} \rangle \chi_{ij} - 3|\langle \chi_{ij} \rangle|^4 \quad (9)$$

$$\Rightarrow \vec{S}_i \cdot \vec{S}_j \approx -2\langle \chi_{ij} \rangle \chi_{ji} + \text{h.c.} + 2|\langle \chi_{ij} \rangle|^2 + \frac{1}{4} \quad (10)$$

$$\Rightarrow \left(\vec{S}_i \cdot \vec{S}_j \right)^2 \approx -\langle \chi_{ij} \rangle \chi_{ji} + 8|\langle \chi_{ij} \rangle|^2 \langle \chi_{ij} \rangle \chi_{ji} + \text{h.c.} + |\langle \chi_{ij} \rangle|^2 - 12|\langle \chi_{ij} \rangle|^4 + \frac{1}{16} \quad (11)$$

Bringing everything together:

$$\begin{aligned}
\mathcal{H} \approx \sum_{\langle ij \rangle} J_{ij} & \left[\left(-1 - \kappa/2 + 4\kappa |\langle \chi_{ij} \rangle|^2 \right) \langle \chi_{ij} \rangle \left(f_{j\uparrow}^\dagger f_{i\uparrow} + f_{j\downarrow}^\dagger f_{i\downarrow} \right) + \text{h.c} \right. \\
& \left. + \left(2 + \kappa - 12\kappa |\langle \chi_{ij} \rangle|^2 \right) |\langle \chi_{ij} \rangle|^2 + \frac{1}{4} + \frac{\kappa}{16} \right]
\end{aligned} \tag{12}$$

It is sometimes handy to define

$$\tilde{J}_{ij} = J_{ij} (1 + \kappa/2), \quad \tilde{\kappa} = \kappa/(1 + \kappa/2) \tag{13}$$

I drop the constant:

$$\begin{aligned}
\mathcal{H} \approx \sum_{\langle ij \rangle} \tilde{J}_{ij} & \left[- \left(1 - 4\tilde{\kappa} |\langle \chi_{ij} \rangle|^2 \right) \langle \chi_{ij} \rangle \left(f_{j\uparrow}^\dagger f_{i\uparrow} + f_{j\downarrow}^\dagger f_{i\downarrow} \right) + \text{h.c} \right. \\
& \left. + 2 \left(1 - 6\tilde{\kappa} |\langle \chi_{ij} \rangle|^2 \right) |\langle \chi_{ij} \rangle|^2 \right]
\end{aligned} \tag{14}$$